Lab 5: FSM –Finite State Machine

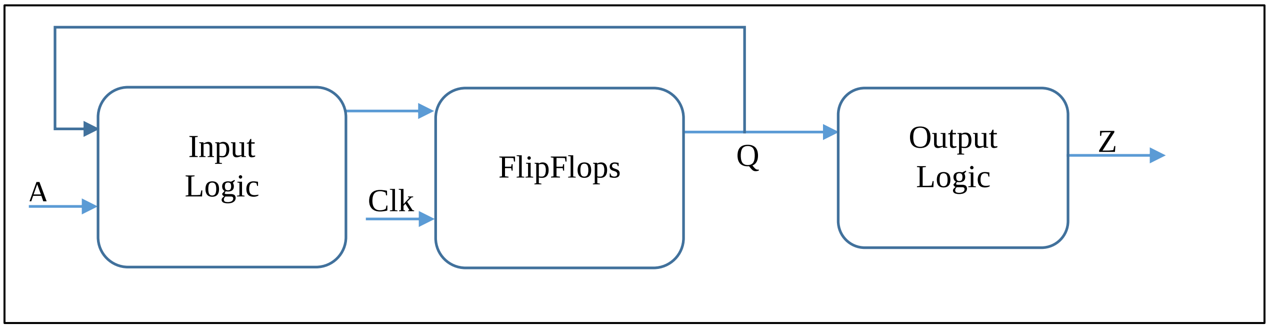


Figure 1: Moore FSM

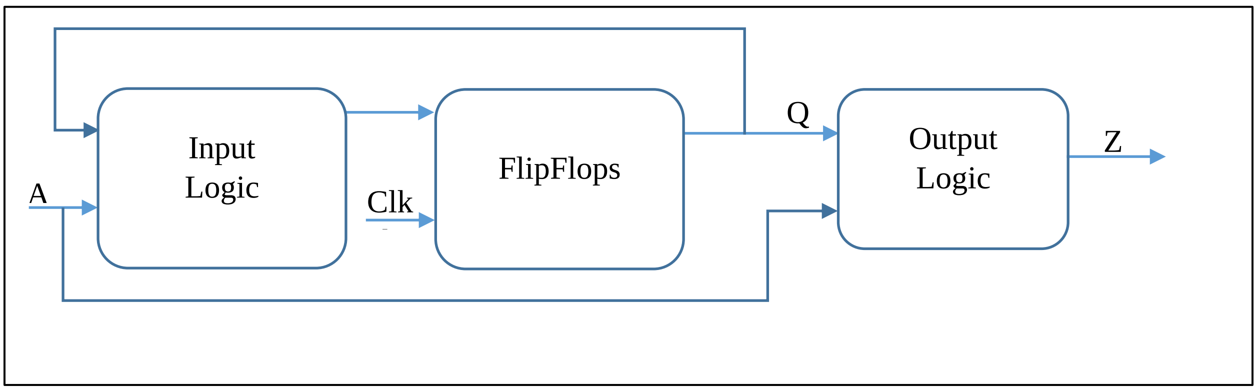


Figure 2: Mealy FSM

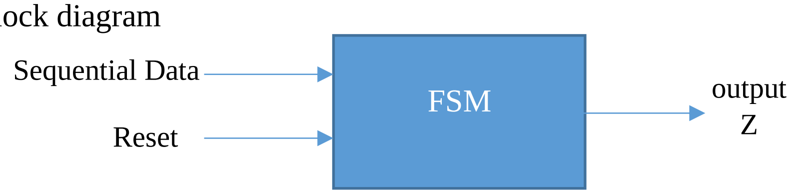
1. Design a Moore FSM that detect the bit squence “101” in a received data stream and accepts overlapping data.

Example:

A=1011010100101

Z=0010010100001

Block diagram



Steps to implement:

a. Contruct the state transition diagram

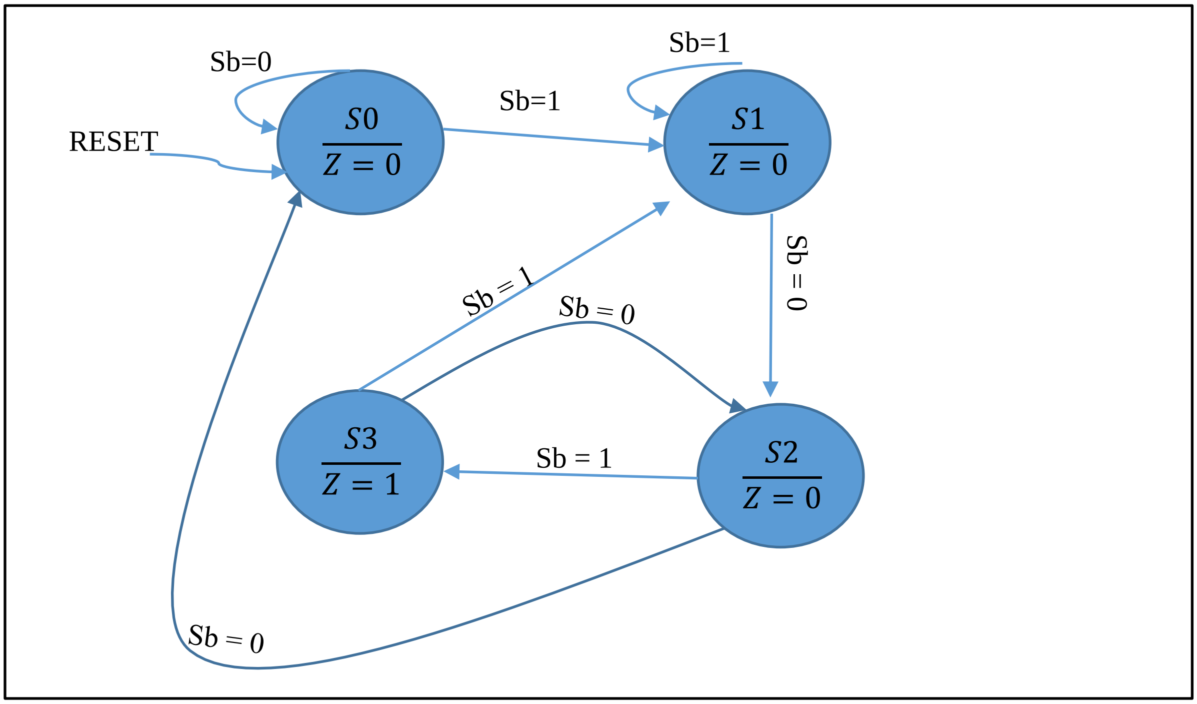


Figure 3: Moore state diagram

b. From the state diagram make the state table

CurrentState	NextState		Output Z
	Sb=0	Sb=1	
S0	S0	S1	0
S1	S2	S1	0
S2	S0	S3	0
S3	S2	S1	1

c. Assign specific state labels.

State	Q1Q0
S0	00
S1	01
S2	10
S3	11

Current state Q1Q0	NextState Q1Q0		Output Z
	Sb=0	Sb=1	
00	00	01	0
01	10	01	0
10	00	11	0
11	10	01	1

d. Make output function that depends on the current state

e. Derive the input functions for the flip-flops based on the output states of the FFs and the input data. we need to formulate the logic that drives the FF inputs. Since we are designing a FSM to detect the squence “101”, we will use the D flip-flops for simplicity, where the input out FF is directly connected to the value of the function we design.

Q1Q0	Q1Q0 Sb=0	Q1Q0 Sb=1	D1D0 Sb=0	D1D0 Sb=1
00	00	01	00	01
01	10	01	10	01
10	00	11	00	11
11	10	01	10	01

f. Build the output functions f(D1) and F(D0) based on Q1, Q0 and the input Sb.

g. Implement the design in Logisim.

2. Design a Mealy FSM for the problem in question 1 using a Mealy FSM.

A Mealy FSM has its output value change immediately when the input value meets the condition. The design process of a Mealy FSM is similar to that of a Moore FSM. The state transistion diagram of the Mealy FSM is presented as follows.

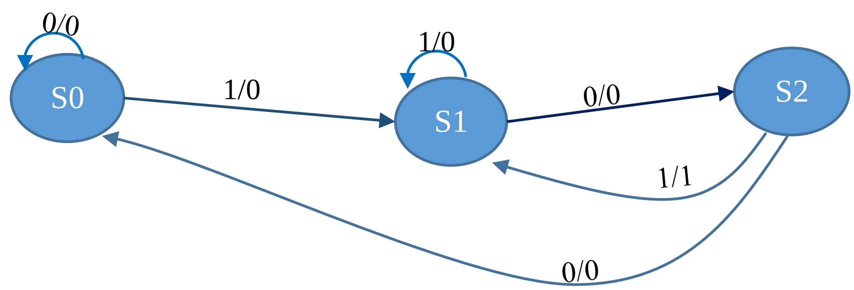


Figure 4: Mealy FSM state diagram

3. Design a Moore FSM for a keyad lock takes 3 BCD numbers as input. If the values of the BCD numbers are 3,4,1 the lock will open (use LEDs to illustrate the input and output)

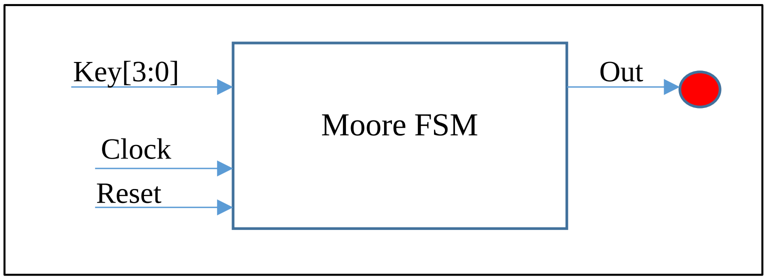


Figure 5: Digital Lock block diagram

4. Design a Mealy FSM with the functionality similar to the problem above, but input values to open the keypad lock are 2,0,2 and 5